



COMP30026

Models of Computation

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Lecture 6

Predicate Logic: Semantics



Expressivity of Predicate Logic

No sorting algorithm can guarantee sorting with fewer than $n \log(n)$ comparisons.



Expressivity of Predicate Logic

No sorting algorithm can guarantee sorting with fewer than $n \log(n)$ comparisons.

$$\forall x (SP(x) \rightarrow \neg(c(x) < n \log n))$$

where:

- $SP(x)$ stands for x is a sorting program (comparison based)
- c is the complexity of a program



Recap: New in Predicate Logic

Five new kinds of symbols:

1. constants

- (e.g. a for “**the** algorithm”, d for “**the** debugger”)
- Usually beginning of alphabet, so $a, b, c, d, \dots$



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5. functions (e.g. $f, g, +$)



Existential Quantification

Existential quantification, $\exists$, is generalised $\vee$.

“Tina found some money and gave **it** to the Red Cross”:

$$\exists x (Money(x) \wedge Found(tina, x) \wedge Gave(tina, x, redcross))$$



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$$\exists x (Money(x) \wedge Found(tina, x) \wedge Gave(tina, x, redcross))$$

means:

$$(Found(tina, \$1) \wedge Gave(tina, \$1, redcross)) \vee$$

$$(Found(tina, \$2) \wedge Gave(tina, \$2, redcross)) \vee$$

$$\vdots$$



Universal Quantification

Universal quantification, $\forall$, is generalised $\wedge$.

“The square of every integer is nonnegative”:

$$\forall x(x \in \mathbb{Z} \rightarrow (x \cdot x \geq 0))$$



Universal Quantification

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“The square of every integer is nonnegative”:

$$\forall x(x \in \mathbb{Z} \rightarrow (x \cdot x \geq 0))$$

means:

$$0 \times 0 \geq 0 \wedge$$

$$1 \times 1 \geq 0 \wedge$$

$$\vdots$$



Quiz: Translate This

Translate “Every program outputs a valid number”.

Use these predicates:

Predicate	Interpretation
$P(x)$	x is a program
$V(x)$	x is a valid number
$O(x, y)$	x output y



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Use these predicates:

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Answer:

$$\forall x(P(x) \rightarrow \exists y(V(y) \wedge O(x, y)))$$

or, equivalently.

$$\forall x \exists y(P(x) \rightarrow (V(y) \wedge O(x, y)))$$



Terms

A **term** represents an **individual** object.

Terms are not formulas.

Examples:

- *redcross*
- 1
- x
- $f(x + y)$



Atomic Formulas

Atomic formulas represent statements about **individual** objects.

Examples:

- $Flies(x)$
- $P(x, c)$
- $Gave(tina, x, redcross)$
- $2 + 2 = 5$



A Notational Convention

A predicate starts with an upper case letter;
nothing else does.

- “*parent(rhonda)*” is a **term**.
 - Likely refers to “the parent of Rhonda”.
- “*Parent(rhonda)*” is a **formula**.
 - Likely represents the proposition “Rhonda is a parent”.



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 - binding precedence $\neg, \forall, \exists > \wedge, \vee > \rightarrow$
 - omit outer parentheses
 - $\rightarrow, \wedge, \vee$ are right-associative

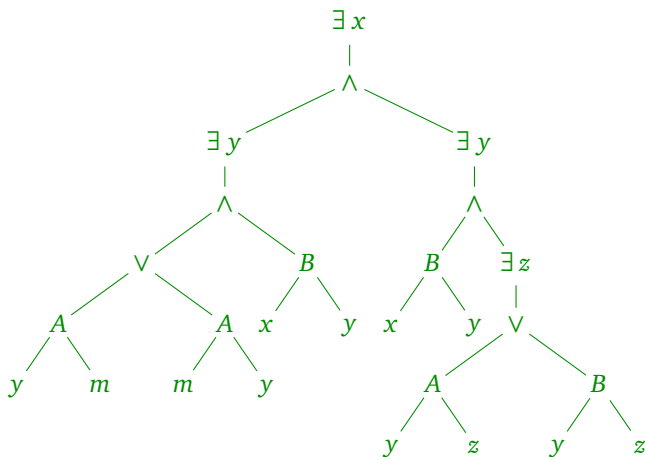


Parse Tree

$\exists x (\exists y ((A(y, m) \vee A(m, y)) \wedge B(x, y)) \wedge \exists y (B(x, y) \wedge \exists z (A(y, z) \vee B(y, z))))$

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Predicates vs Functions and their types

	take as input	produce
connectives $\wedge, \rightarrow, \dots$	propositions	a proposition
predicates $P(..), =, \dots$	objects	a proposition
functions $f, +, \dots$	objects	an object



Quantifier Scope

The subformula attached to a quantifier is its **scope**.

Example: $\forall x (P(x) \vee Q(c))$
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If a variable is not bound, it is **free**.

Example: x is free in $P(x)$.



Poll

Which variables are bound, and which are free?

1. $\forall x(P(x) \rightarrow Q(y))$
2. $\forall z(P(x, y, z) \wedge \forall y(P(f(x), z, y)))$
3. $\forall x \exists y(x < y \wedge \exists z(y < z))$



Renaming Variables

Bound variables can be renamed unless there is a clash:

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Renaming **free** variables changes meaning:

$P(x)$ is different to $P(y)$.



From English to Predicate Logic

A rough guide:

1. Identify nouns, verbs, pronouns, adjectives, relative clauses.
2. Assign:
 - Constant symbols to singular objects (e.g. “Rhonda”).
 - Predicate symbols to verbs and adjectives (e.g. “loves”).
 - Function symbols to relative clauses (e.g. “the parent of”).
 - Variables to indefinite pronouns (e.g. “someone”).
3. Replicate logical structure of sentence.



Example Translations

Let $L(x, y)$ stand for “x loves y”

Alex loves Eva

Everyone loves Eva (incl. Eva)

Eva is loved by everyone else

Someone other than Alex loves
Alex

Everybody loves somebody

Someone is loved by everybody

Someone loves everybody



Example Translations

Let $L(x, y)$ stand for “x loves y”

$L(alex, eva)$

Alex loves Eva

$\forall x L(x, eva)$

Everyone loves Eva (incl. Eva)

$\forall x (\neg(x = eva) \rightarrow L(x, eva))$

Eva is loved by everyone else

$\exists x (\neg(x = alex) \wedge L(x, alex))$

Someone other than Alex loves Alex

$\forall x \exists y L(x, y)$

Everybody loves somebody

$\exists y \forall x L(x, y)$

Someone is loved by everybody

$\exists x \forall y L(x, y)$

Someone loves everybody



Word Order

Consider word order with care:

- “There is something which is not P ”: $\exists y \neg P(y)$
- “There is not something which is P ” (“nothing is P ”):
 $\neg \exists y P(y)$
- “All S are not P ” **vs** “not all S are P ”:
 $\forall x (S(x) \rightarrow \neg P(x))$ **vs** $\neg \forall x (S(x) \rightarrow P(x))$



Quantifier Order

Order of different quantifiers is important!!!

$\forall x \exists y L(x, y)$ says “everyone has someone they love”.

$\exists y \forall x L(x, y)$ says “there is someone who is loved by everyone”.



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$\forall x \exists y L(x, y)$ says “everyone has someone they love”.

$\exists y \forall x L(x, y)$ says “there is someone who is loved by everyone”.

But $\forall x \forall y$ is the same as $\forall y \forall x$ and $\exists x \exists y$ is the same as $\exists y \exists x$.



Implicit Quantifiers

Often quantifiers are implicit in English.

Look for nouns (especially plural) without determiners.

“Humans are mortal” means “**all** humans are mortal”:

$$\forall x(Human(x) \rightarrow Mortal(x))$$

“A horse is stronger than a dog” would usually mean:

$$\forall x \forall y((Horse(x) \wedge Dog(y)) \rightarrow Stronger(x, y))$$

“If a child owns a dog, the child spoils it”:

$$\forall x \forall y((Child(x) \wedge Dog(y) \wedge Owns(x, y)) \rightarrow Spoils(x, y))$$



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True or false?

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2. **True** if discussing $\mathbb{Z}$ and $<$ is “less than or equal”.



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3. **True** if discussing $\mathbb{R}$ and $<$ is “less than”.



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Depends on how we interpret it.

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2. **True** if discussing $\mathbb{Z}$ and $<$ is “less than or equal”.
3. **True** if discussing $\mathbb{R}$ and $<$ is “less than”.
4. Unconditionally **true** if universe is $\{0\}$.